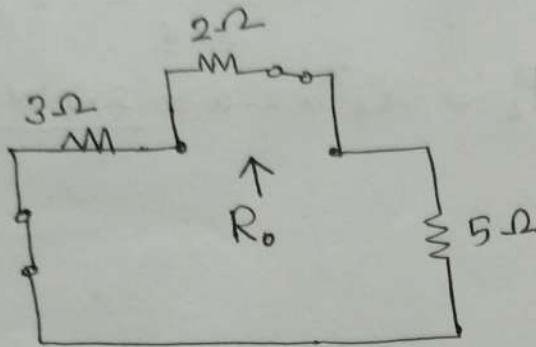


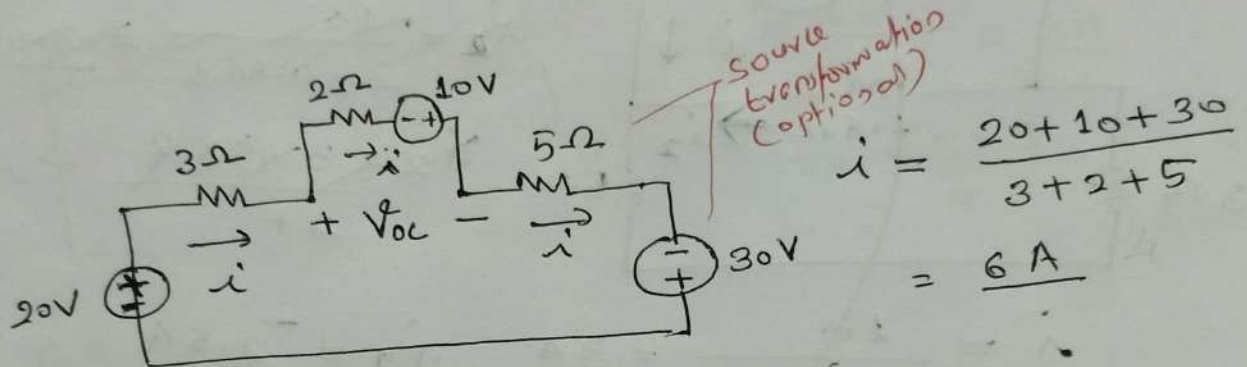
EEE FIII Electrical Sciences
Midsemester Exam Solutions

Q1

(A)



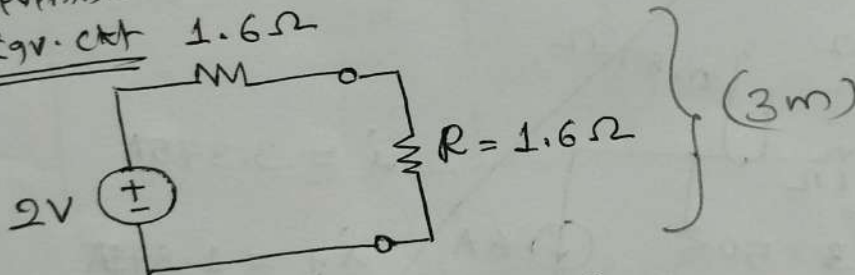
$$R_0 = 8 \parallel 2 = \underline{1.6 \Omega} \quad (4m)$$



$$i = \frac{20 + 10 + 30}{3 + 2 + 5} = \underline{6 A}$$

$$\therefore V_{oc} = 2i + 10 = \underline{2 V} \quad (4m)$$

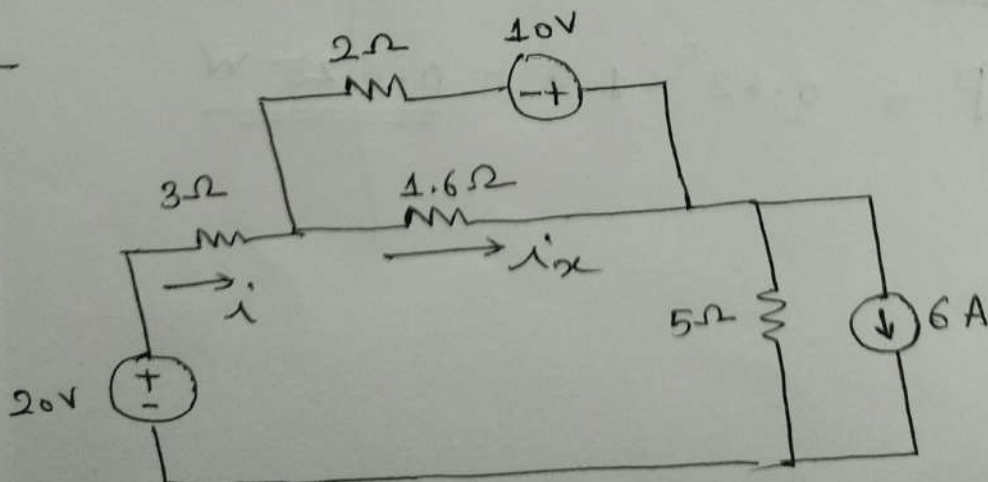
Therminols
Egv. ckt

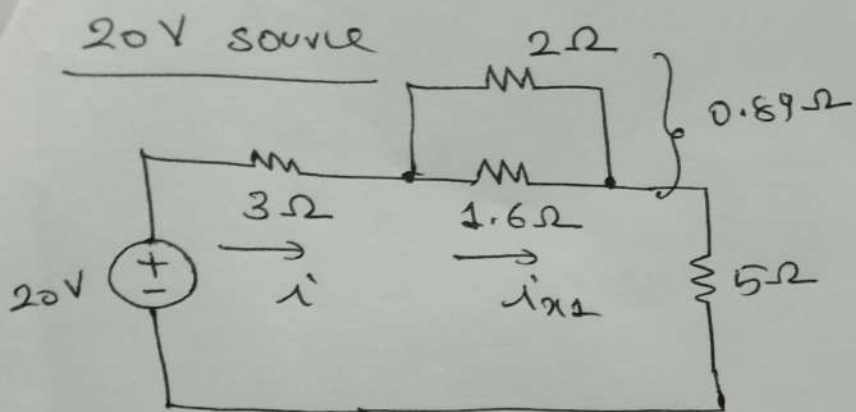


$$P_{max} = \left(\frac{2}{2 \times 1.6} \right)^2 \times 1.6 = \underline{\underline{0.625 W}} \quad (4m)$$

Q1

(B)

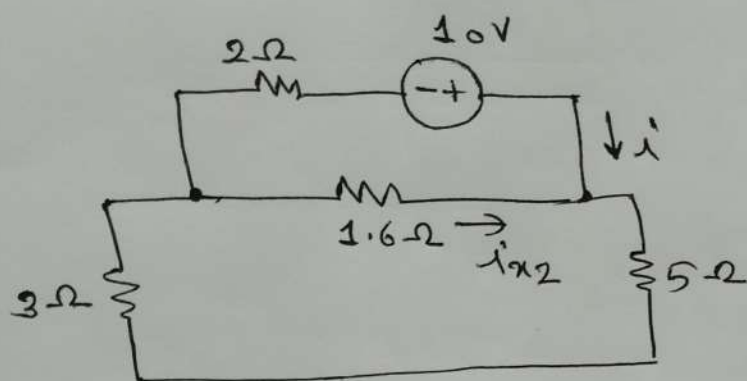




$$i = \frac{20}{8.89} = \underline{2.25 \text{ A}}$$

$$i_{x1} = \frac{2.25 \times 2}{3.6} = \underline{1.25 \text{ A}} \quad (4m)$$

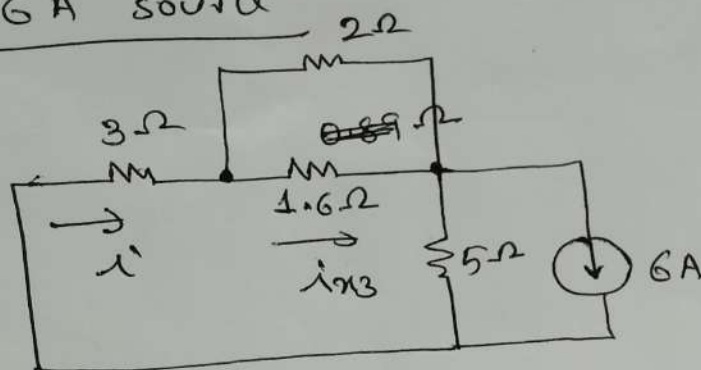
10V source



$$i = \frac{10}{2 + 1.33} = \underline{3 \text{ A}}$$

$$i_{x2} = \frac{3 \times 8}{9.6} = \underline{2.5 \text{ A}} \quad (4m)$$

6A source



$$i = \frac{6 \times 5}{8.89} = \underline{3.375 \text{ A}} \quad (4m)$$

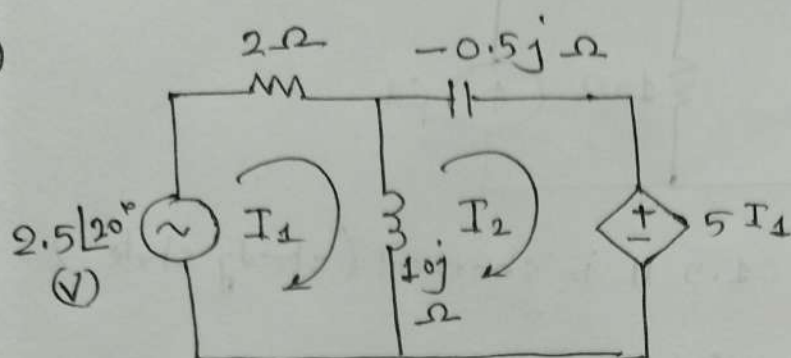
$$i_{x3} = \underline{1.875 \text{ A}}$$

$$\therefore i_x = i_{x1} + i_{x2} + i_{x3} = \underline{0.625 \text{ A}}$$

$$P = i_x^2 \times 1.6 = \underline{\underline{0.625 \text{ W}}} \quad (3m)$$

Q2

(A)



frequency domain circuit

(3m)

if all elements correct
else zero.

$$\underline{M1} : 2.5 \angle 20^\circ - 2I_1 - 10j(I_1 - I_2) = 0$$

$$(2 + 10j)I_1 - 10jI_2 = 2.5 \angle 20^\circ \rightarrow \textcircled{1}$$

(2m)

$$\underline{M2} : 10j(I_1 - I_2) + 0.5jI_2 = 5I_1$$

$$10jI_1 - 9.5jI_2 = 5I_1$$

$$(-5 + 10j)I_1 - 9.5jI_2 = 0 \rightarrow \textcircled{2}$$

(2m)

Solving ① and ②

$$I_1 = \frac{2.5 \angle 20^\circ}{[2 + 10j - 10j(-5 + 10j)/9.5j]}$$

$$= \underline{0.343 \angle 24.14^\circ \text{ (A)}}$$

$$I_2 = \underline{0.403 \angle 50.7^\circ \text{ (A)}}$$

$$\therefore i_1(t) = 0.343 \cos(10t + 24.14^\circ) \text{ (A) (3m)}$$

$$i_2(t) = 0.403 \cos(10t + 50.7^\circ) \text{ (A) (3m)}$$

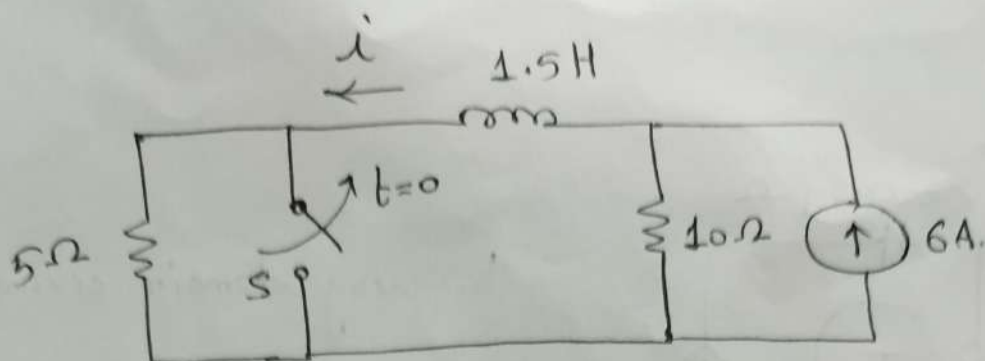
Avg. power absorbed by the dependent source

$$P_{av} = \text{Re} \left\{ \frac{5I_1 \times I_2^*}{2} \right\} = \underline{0.309 \text{ (W)}}$$

(2m)

(Q2)

(B)



$t < 0$: S is closed, 1.5 H is shorted (steady state)

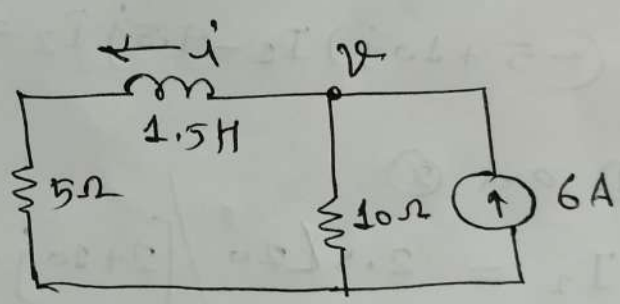
$$\therefore i(t) = \underline{\underline{6A}} \quad (3m)$$

$t = 0$: $i(0) = i(0^-) = \underline{\underline{6A}} \quad (3m)$

$t \rightarrow \infty$ S is open, 1.5 H is again shorted

$$i(\infty) = \frac{6 \times 10}{15} = \underline{\underline{4A}} \quad (3m)$$

$t > 0$



$$6 = \frac{v}{10} + i, \quad v = 1.5 \frac{di}{dt} + 5i$$

$$\therefore 6 = 0.15 \frac{di}{dt} + 0.5i + i$$

$$\frac{di}{dt} + 10i = 40$$

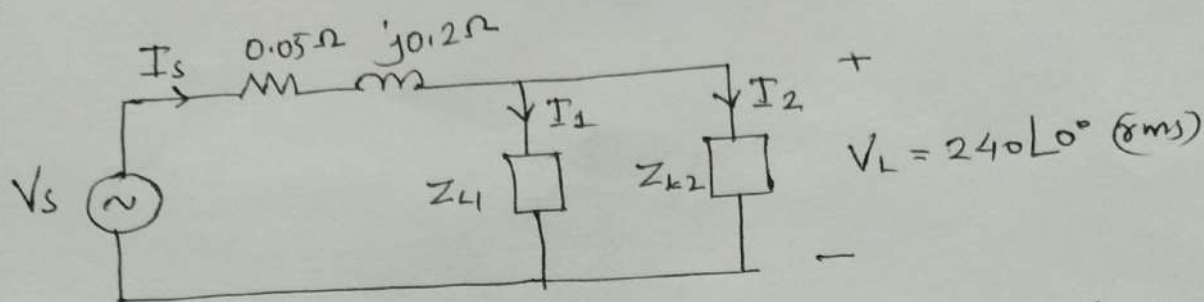
$$i(t) = A e^{-10t} + i(\infty)$$

$$= A e^{-10t} + 4 \Rightarrow A = 2 \text{ for } i(0) = 6$$

$$\therefore i(t) = \underline{\underline{4 + 2e^{-10t}}} \quad (A), \quad t > 0 \quad (6m)$$

Q3

(A)



$$I_1 = \frac{20k}{0.7 \times 240} \left[+\cos^{-1}(0.7) \right] = \underline{119 \angle 45.6^\circ \text{ (A)}} \quad (2)$$

$$I_2 = \frac{12k}{0.9 \times 240} \left[-\cos^{-1}(0.9) \right] = \underline{55.6 \angle -25.8^\circ \text{ (A)}} \quad (2)$$

$$(a) \therefore I_s = I_1 + I_2 = \underline{146.5 \angle 24.5^\circ \text{ (A)}} \quad (2)$$

$$(b) V_s = I_s (0.05 + j0.2) + 240 \angle 0^\circ$$

$$= \underline{236.4 \angle 7.2^\circ \text{ (V)}} \quad (3)$$

$$(c) S = V_s I_s^* = \underline{3460 \angle -17.3^\circ \text{ (VA)}} \quad (1)$$

$$(d) \text{pf} = \cos(7.2 - 24.5) = \underline{0.955 \text{ leading}} \quad (1)$$

(Q3)

B

$$\cancel{Q} = \frac{\cancel{BW}}{\cancel{f_r}} \Rightarrow Q = \frac{f_r}{BW} = \underline{10} \quad (2)$$

$$Q = \omega_r R C \Rightarrow C = \frac{Q}{\omega_r R} = \underline{159 \text{ pF}} \quad (2)$$

$$\omega_r = \frac{1}{\sqrt{LC}} \Rightarrow L = \frac{1}{\omega_r^2 C} = \underline{159 \mu\text{H}} \quad (2)$$

$$\text{At resonance: } I_R = I_0 = \underline{0.001 \angle 0^\circ \text{ A}} \quad (3)$$

$$I_C = \frac{V_R}{(1/j\omega_r C)} = \frac{(I_R \times R) \times j\omega_r C}{1} = \underline{0.01j \text{ (A)}} \quad (3)$$

$$I_L = \frac{I_R \times R}{j\omega_r L} = \underline{-0.01j} = \underline{-I_C} \quad (3)$$